

YUVRAJ PAWAR

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PROFESSIONAL SUMMARY

Computer Science undergraduate who enjoys breaking down complex technical problems and understanding how things work under the hood. Actively expanding my expertise in full-stack development, machine learning, and DSA. Looking for an internship opportunity to contribute to hands-on engineering projects.

TECHNICAL SKILLS

Languages: JavaScript, Python, C++, Java

Frontend: React, HTML5, CSS3, Tailwind

Backend: Node.js, Express, Flask

Databases: MongoDB, PostgreSQL, MySQL

Core CS: Data Structures & Algorithms, Operating Systems, Computer Networks, Network Security

Tools: Git, GitHub, REST APIs, Postman, Render

PROJECTS

EcoSimple – Environmental Monitoring Platform

React, Flask, MySQL, Leaflet, Recharts, OpenWeatherMap API

- Built a full-stack environmental monitoring platform to visualize AQI and water quality metrics.
- Integrated OpenWeatherMap API data for 10+ cities with interactive pollution heatmaps using Leaflet.
- Created dashboards using Recharts to analyze pollution trends and pollutant comparisons.

Answer Sheet Rechecking Tool

Python, OpenCV, OCR, Groq LLM, Node.js, React, MongoDB

- Architected an AI-assisted evaluation system enabling students to upload answer sheets for automated pre-checking.
- Implemented OCR pipelines with OpenCV and integrated Groq LLM for intelligent error detection and feedback generation.
- Optimized backend APIs for faster processing and structured evaluation reports.

Generic Medicine Recommender

Node.js, React, MongoDB, Tesseract OCR, Groq LLM

- Developed an OCR-driven system to extract medicine names from prescription images.
- Integrated AI-based text analysis to recommend cost-effective generic alternatives.
- Designed RESTful APIs and structured MongoDB schemas for efficient query performance.

Railway Track Fault Detection Research

YOLOv8, ResNet18, PyTorch, OpenCV

- Conducted a comparative study of classification and object detection approaches for automated railway track fault detection.
- Implemented ResNet18 for fault classification and YOLOv8 for real-time defect localization using bounding boxes.
- Improved model robustness using data augmentation techniques including flipping, brightness variation, and noise injection.
- Developed a pseudo-labeling pipeline to leverage unlabeled datasets and improve cross-dataset generalization.
- Evaluated model performance using precision, recall, and mAP metrics on railway and structural crack datasets.

EDUCATION

Diploma in Computer Science

2023 – 2026

Dr. Vishwanath Karad MIT World Peace University

Data Structures · DBMS · Operating Systems · Computer Networks · Software Engineering

B.Tech in Computer Science

2026 – 2029